

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering
V Semester - CSE 'A'
UCS2511 NETWORKS LAB
Exercise 10
DNS Resolution using UDP

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Aim:

DNS Resolution using UDP

server.c

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` Simulation of DNS in socket programming in C

server.c

```
// server program for udp connection
#include <stdio.h>
#include <strings.h>
#include <stdlib.h>
```

```
#include <sys/types.h>
```

```

#include <arpa/inet.h>
#include <sys/socket.h>
#include <netinet/in.h>
#include <string.h>

#define PORT 5000
#define MAXLINE 1000
#define MAX 10

typedef struct
{

    char * domain_name;

    char * IP;
} dom;

dom domain_arr[MAX];

// Driver code
int main()
{

    char buffer[100];
    char *message = "Server connected";
    int listenfd, len;
    struct sockaddr_in servaddr, cliaddr;
    bzero(&servaddr, sizeof(servaddr));

    // Create a UDP Socket
    listenfd = socket(AF_INET, SOCK_DGRAM, 0);
    servaddr.sin_addr.s_addr = htonl(INADDR_ANY);
    servaddr.sin_port = htons(PORT);
    servaddr.sin_family = AF_INET;

    //reqd ip
    char *correct = (char*)malloc(MAXLINE);

```

```

char *modify = (char*)malloc(MAXLINE);
char *new_ip = (char*)malloc(MAXLINE);
strcpy(correct,"IP not found");
//create table
domain_arr[0].domain_name = "youtube.com";


domain_arr[0].IP = "1.2.3.4";
domain_arr[1].domain_name = "google.com";
domain_arr[1].IP = "6.7.8.9";
domain_arr[2].domain_name = "chatgpt.ai";
domain_arr[2].IP = "10.11.12.13";


for(int i=0;i<3;i++)
{

printf("IP %s Domain name = %s \n",domain_arr[i].IP,
domain_arr[i].domain_name);


}
// bind server address to socket descriptor
bind(listenfd, (struct sockaddr*)&servaddr,

sizeof(servaddr));


//receive the datagram
len = sizeof(cliaddr);
int n = recvfrom(listenfd, buffer, sizeof(buffer),

0, (struct sockaddr*)&cliaddr,&len); //receive
message from client


printf("Buffer is %s ",buffer);
/*
int n1 = recvfrom(listenfd, modify, sizeof(buffer),

```

```
0, (struct sockaddr*)&cliaddr,&len);  
printf("Modify = ? %s \n ",modify);*/
```

```
printf("Want to modify? yes or no ");  
fgets(modify,MAXLINE,stdin);  
printf("The value is %d ",strncmp(modify,"yes",3));  
if(strncmp(modify,"yes",3)==0)  
{
```

```
printf("Enter IP to replace. Will replace if found
```

```
");
```

```
    //printf("Result %d \n",strncmp(domain_arr[i].domain_name,buffer,sizeof(domain_arr[0  
]));
```

```
if(strncmp(domain_arr[i].domain_name,buffer,sizeof(domain_arr[0  
]))+10 == 0)
```

```
{  
    printf("Found the Required IP  %s  Domain name
```

```
= %s ",domain_arr[i].IP, domain_arr[i].domain_name);  
    if(strncmp(modify,"yes",3)==0)
```

```
{  
    domain_arr[i].IP = new_ip;
```

```
}  
strcpy(correct,domain_arr[i].IP);
```

```
}}
```

```
buffer[n] = '\0';  
puts(buffer);
```

```
// send the response  
sendto(listenfd, correct, MAXLINE, 0,
```

```
(struct sockaddr*)&cliaddr, sizeof(cliaddr));
```

```
if(strncmp(modify,"yes",3)==0)  
{
```

```
    sendto(listenfd, new_ip, MAXLINE, 0,
```

```
    (struct sockaddr*)&cliaddr, sizeof(cliaddr));  
}
```

```
for(int i=0;i<3;i++)  
{
```

```
    printf("IP  %s  Domain name = %s \n",domain_arr[i].IP,  
domain_arr[i].domain_name);
```

```
}}
```

```
//client.c
```

```
// udp client driver program
```

```
#include <stdio.h>
```

```
#include <strings.h>
```

```
#include <string.h>
```

```
#include <sys/types.h>
```

```
#include <arpa/inet.h>
```

```
#include <sys/socket.h>
```

```
#include<netinet/in.h>
```

```
#include<unistd.h>
```

```
#include<stdlib.h>
```

```
#define PORT 5000
```

```
#define MAXLINE 100
```

```
// Driver code
```

```
int main()
```

```
{
```

```
    char buffer[100];
```

```
    char *message = (char *)malloc(MAXLINE);
```

```
    printf("Enter domain name : ");
```

```
    fgets(message,MAXLINE,stdin);
```

```
    int sockfd, n;
```

```
    struct sockaddr_in servaddr;
```

```
// clear servaddr
```

```
    bzero(&servaddr, sizeof(servaddr));
```

```
    servaddr.sin_addr.s_addr = inet_addr("127.0.0.1");
```

```

servaddr.sin_port = htons(PORT);
servaddr.sin_family = AF_INET;


// create datagram socket
sockfd = socket(AF_INET, SOCK_DGRAM, 0);


// connect to server


if(connect(sockfd, (struct sockaddr *)&servaddr,
sizeof(servaddr)) < 0)

{
    printf("\n Error : Connect Failed \n");
    exit(0);
}


// request to send datagram
// no need to specify server address in sendto
// connect stores the peers IP and port


sendto(sockfd, message, MAXLINE, 0, (struct
sockaddr*)NULL, sizeof(servaddr));


/*printf("Do you want to update the ip? yes or no: ");
char *x = (char *)malloc(MAXLINE);
fgets(x,MAXLINE,stdin);
printf("\n Sending %s ",x );


sendto(sockfd, x, MAXLINE, 0, (struct sockaddr*)NULL,
sizeof(servaddr));


if(strncmp(x,"yes",3)!=0)
{
    char *x1 = (char *)malloc(MAXLINE);
    printf("Enter the new IP. Will update it if found");
    fgets(x1,MAXLINE,stdin);
    sendto(sockfd, x1, MAXLINE, 0, (struct sockaddr*)NULL,

```

```
sizeof(servaddr));
    }*/

    // waiting for response

    recvfrom(sockfd, buffer, sizeof(buffer), 0, (struct
sockaddr*)NULL, NULL);

    printf("The IP of the domain is %s \n" ,buffer);

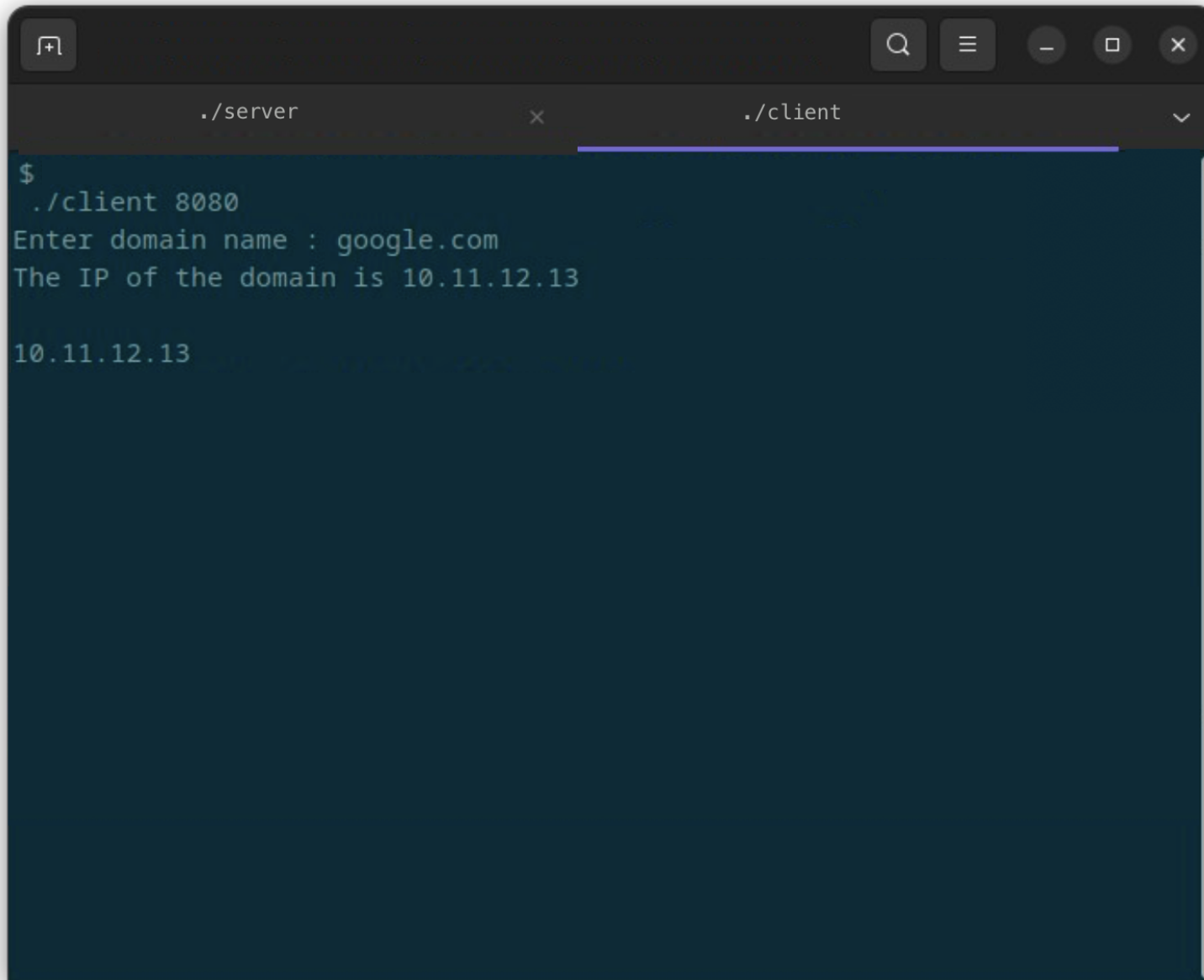
    recvfrom(sockfd, buffer, sizeof(buffer), 0, (struct
sockaddr*)NULL, NULL);

    printf("%s" ,buffer);

    // close the descriptor

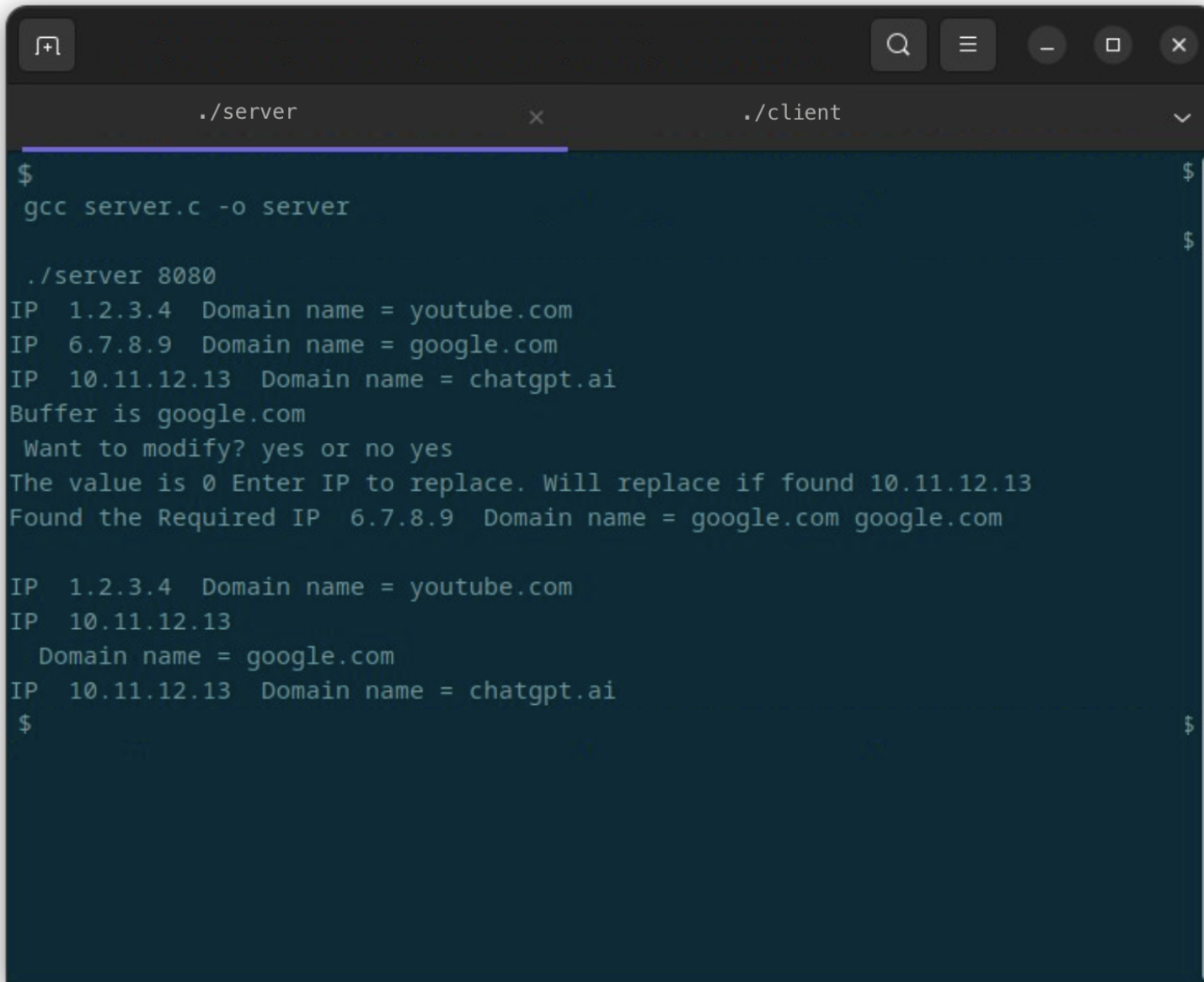
    close(sockfd);
}
```

O/P:



A terminal window with a dark theme. The title bar at the top shows standard window controls (minimize, maximize, close) and a search icon. Below the title bar, there are two tabs: `./server` and `./client`, with the `./client` tab selected. The terminal content shows the following sequence of commands and output:

```
$  
./client 8080  
Enter domain name : google.com  
The IP of the domain is 10.11.12.13  
10.11.12.13
```

A terminal window with two tabs: './server' and './client'. The './server' tab is active and shows the following commands and output:

```
$  
gcc server.c -o server  
  
./server 8080  
IP 1.2.3.4 Domain name = youtube.com  
IP 6.7.8.9 Domain name = google.com  
IP 10.11.12.13 Domain name = chatgpt.ai  
Buffer is google.com  
Want to modify? yes or no yes  
The value is 0 Enter IP to replace. Will replace if found 10.11.12.13  
Found the Required IP 6.7.8.9 Domain name = google.com google.com  
  
IP 1.2.3.4 Domain name = youtube.com  
IP 10.11.12.13  
Domain name = google.com  
IP 10.11.12.13 Domain name = chatgpt.ai  
$
```

Learning Outcomes:

I learned how to implement flow control using socket programming.